SmartSchedule — Improvement Plan (short version)

For developers: what we build next, how it should look, and what to decide before coding.

28 Oct 2026 · Draft for team review · Full plan: `Improve_Plan/improve_plan.md` · Engine reference: `Baseline_Plan/baseline_plan.md`

1. **Build a student-facing Week Planner on top of the existing engine.** The baseline benchmark workbench stays, but moves behind a **Lab mode** switch for testing, grading and the demo. Same engine, two faces.
2. **Design: calm, minimal, explainable.** One primary action per screen. Colour shows block type only, and red means conflicts only. Every automatic block has **"Why here?"**, and every automatic change has a **preview + Undo**.
3. **Start the UI now on mock JSON (W1), not in W6–W7.** Before engine code, freeze one rule sheet: our documents still use different values for breaks, sleep, priority and urgency (see "Decisions", page 4).

1 · What changes

Student app new

- Week calendar with normal 30-min rows and sleep folded into one bar. The 672-slot grid stays in the code.
- Add tasks fast (form or Quick-add syntax), press **Plan week**, review the proposal, accept.
- Drag, pin, undo; "Something came up" repairs only what is hit; overload is explained in plain numbers.

Lab mode kept from baseline §7

- Scenario loader T01–T10, algorithm selector (EDF, LST, scored greedy + repair, backtracking N ≤ 8).
- KPIs (hard violations, deadline rate, ms, stability, preference fit), side-by-side compare, local vs. full rebuild.
- **Determinism check:** run 3× and compare the output hash. Direct evidence for the CT criterion.

2 · Design direction — modern, minimalist, clean, easy

P1 **One screen, one job.** One primary button per screen.

P2 **Show the plan, hide the machinery.** No slots, solver names or KPIs outside Lab.

P3 **Ask little, suggest safely, always confirm.** Only 3 required fields: title, duration, due.

P4 **Explain every automatic decision.** Text built from engine reason codes.

P5 **Never surprise.** Preview first, Undo always, pinned blocks never move.

P6 **Be honest when it doesn't fit.** Shortage in minutes, options least harm first.

P7 **Colour carries meaning.** Red only for conflicts; every colour also has an icon or shape.

P8 **Accessible, bilingual.** WCAG AA, keyboard, 44-px targets, 375-px width, VI/EN.

Tokens

bg #FAFAFA · surface · text #18181B · muted · accent #4F46E5 · accent-soft · fixed · habit · ok · warn · danger

Font **Inter** (or Be Vietnam Pro), tabular numbers · sizes 12/14/16/20/24/32 · weights 400/500/600 · spacing 4-px base · corners 6/10/16 px · borders for structure, shadows only on overlays · Lucide icons · 150–200 ms motion, moved blocks slide old → new · dark mode from the same tokens.

Block types (replaces the baseline §7.6 colours)

Lecture 🔒	Fixed: class, lab, job, meal	Sleep	Folded bar, protected	● Essay	Deep task
◐ Review	Medium task	○ Submit	Light task	↻ Gym	Habit (soft)
● Proposed	Dashed until Accept	📌 Pinned	Never moved by re-plan	at risk · not planned	Badges, not red blocks

3 · Two solutions for how students use the app

Both use the same engine and data. They differ in what the student sees first and how often they interact.

A · Week Planner recommended MVP

Idea: "my week on one page". Plan in a batch on Sunday/Monday, adjust when things change.

Flow: set up sleep + timetable once → add tasks anytime → **Plan week** → dashed proposal → Accept → drag/pin → "Something came up" → change review (old → new) → Keep / Undo.

Best at: showing the whole algorithm (demo, grading); 1:1 with the 672 grid and T01–T10; low risk.

Weak at: crowded on phones; no natural moment to collect actual durations, so the learning loop gets no data.

B · Today Coach phase 2 screens

Idea: "what should I do now?" One **Now card** (Start / Done / Snooze / Skip); the week is secondary. Mobile-first.

Flow: morning agenda → Now card → Done / Partly / Need more time (+15/+30/+60) → ripple re-plan → 30-s evening check-in → weekly recap with *suggested* profile changes [Apply].

Best at: daily use; produces the real data Pillar 2 needs (actual time, completion by hour); built-in study instruments.

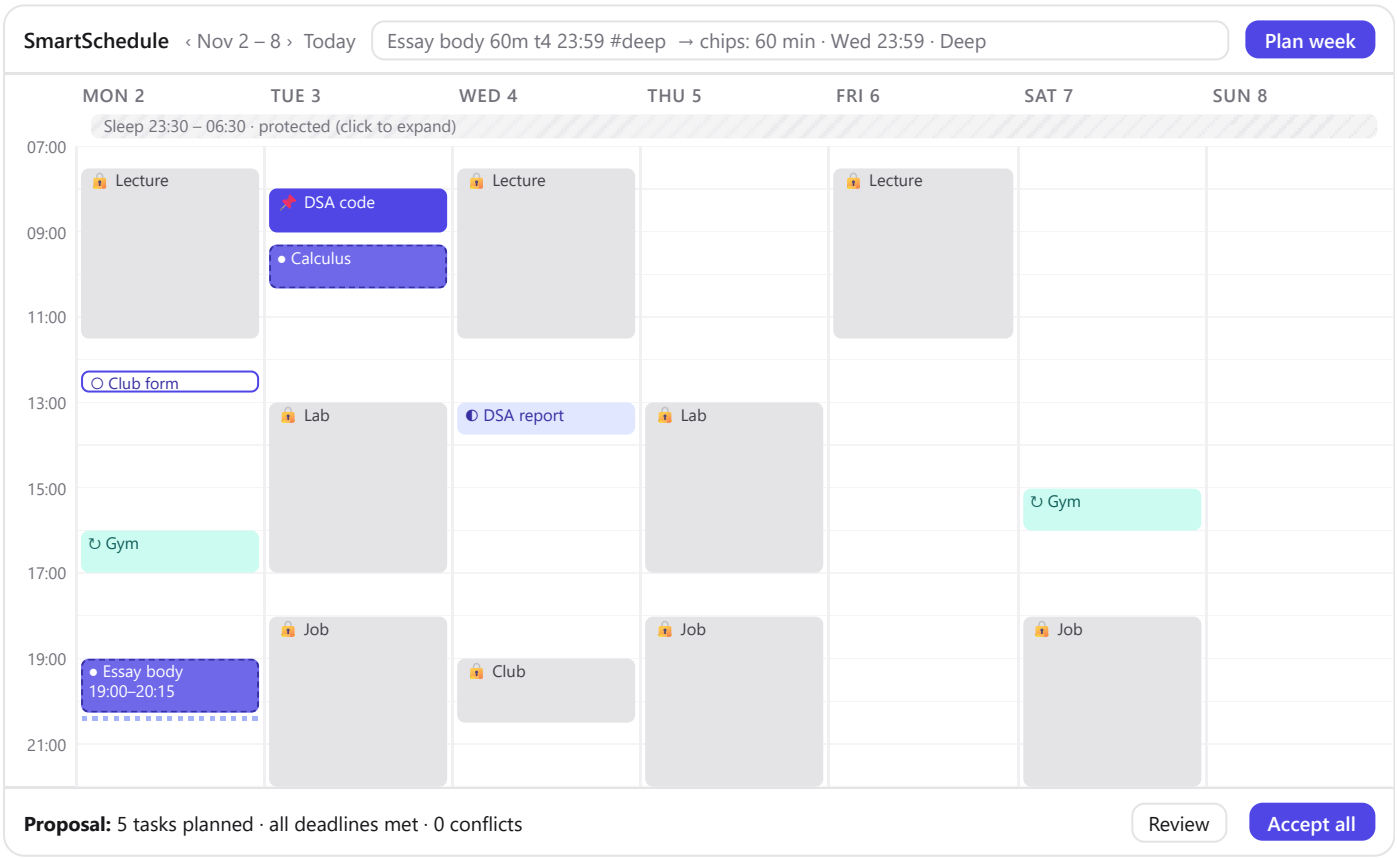
Weak at: hides the week view; needs hosting, persistence and notifications; harder to demo.

Recommendation — "A first, then B's loop". Build Solution A as the MVP (W1–W7) together with Lab mode. Then add B's **Today** agenda and **evening check-in** (cheap, same API), but only if the team confirms the 7-day user study (decision D12, by W3).

Phase 1 — MVP
Onboarding · Week · Tasks · TaskSheet · Why here? · Pin/Drag/Undo · Something came up · Overload · Lab

Phase 2 — if study = yes
Today + Now card · Progress / check-in · Weekly recap (suggestions) · PWA install · Dark mode

4 · Main screen sketch (Lan's busy week, Solution A)



Illustrative layout only, using placements from the Pillar 4 worked example (times rounded to the 15-min grid). Dashed = proposed, not yet accepted; dotted line = 15-min break after Deep work. Click a block → "Why here?": before deadline · high-focus hours · break booked · planned 1st of 5.

5 · MVP feature list

ID	FEATURE	PRIORITY	ENGINE / PILLAR LINK	TESTS
F01	Sleep window (hard lock), set in onboarding	**Must**	Sleep rule	all
F02	Weekly timetable: recurring + one-off events (form, or paint on the grid)	**Must**	Fixed events (Pillar 1 B1)	T01
F06	**TaskSheet**: title, duration (±15 stepper), due (date + time / none), **counts for** (grade weight → priority), focus Deep/Medium/Light (suggested), split + min session, after-task, type	**Must**	Task model T_i (Pillar 3)	T01–T03, T08
F09	**Plan week** with proposal mode (dashed → Accept / Discard)	**Must**	Stages 0–3 / recipe Steps 1–3	T01, T02
F10	**Why here?** popover built from engine reason codes	**Must**	Transparency principle	all
F12	**Pin** a block (treated as fixed by re-plan and repair)	**Must**	Stage 4	T04
F13	**Something came up** → local repair → change review (old → new) + Undo; "rebuild whole week" in advanced	**Must**	Stage 4 ripple / Step 4	T04
F15	**Overload panel**: "Not enough time" vs. "No schedule found"; options least harm first with Preview; sleep never offered	**Must**	OVERLOAD / Step 5	T06, T02
F19	**Lab mode**: scenarios, algorithm selector, KPIs, compare, determinism check	**Must**	Baseline workbench	T01–T10
F03	Import timetable from `.ics` (check first if the FIT portal exports it)	**Should**	Master timetable	edge 1
F04	3 onboarding questions: focus time · estimate habit · disturbed hours (skippable)	**Should**	Pillar 2 Part C (cold start)	T09
F05	Quick-add syntax: `90m` · `t4 13:00` · `next mon` · `w30%` / `p5` · `#deep` · `~45` / `nosplit` · `after:PRJ-1`. Parsed chips shown before saving	**Should**	Deterministic stand-in for M1	T07
F07	Task types: Assignment / Exam prep (spaced D−7, −5, −3, −1) / Project / Habit (×/week) / Errand	**Should**	Pillar 2 deadline types	T02
F08	Break a task into steps (student writes the steps; never split blindly)	**Should**	Pillar 1 A1; report §7.8.2	T03
F11	Drag to move: 15-min snap, allowed zones shaded, blocked drop shows the reason	**Should**	Allowed set from Stage 2	T04
F14	Progress: Done / Partly (%) / Need more time	**Should**	UC9	T05
F16	Deadlines after this week ("due next Mon 09:00")	**Should**	2-week horizon	T10
F17–18	Undo / redo (20 steps) · export `.ics` /JSON/CSV · backup / restore	**Should**	P5 never surprise	—
F20	Vietnamese / English (dates, reason sentences)	**Should**	Target users	—
F21	Natural-language add (M1). Only fills the **same TaskSheet**; never writes to the schedule directly	**Should***	M1, course requirement R4	T07
F22–25	Today + Now card · evening check-in · weekly recap as suggestions · dark mode	**Could**	Phase 2 (Solution B)	T05, T09

* Must, if the course requires an AI component. **Not this semester:** native mobile app, voice input, two-way Google Calendar sync, shared calendars, syllabus PDF reading, silent auto-learning, per-slot location context.

6 · Decisions to make before coding

#	DECISION	WHAT OUR DOCUMENTS SAY	RECOMMENDED DEFAULT
D1	Running example	Minh (report) · Lan (pillars)	Keep both: Minh = calm week, Lan = busy week
D2	Break after Deep work	15 min (report) · 10 min (pillars)	**15 min = 1 slot** (10 min can't exist on the grid)
D3	Break: hard or soft?	Soft S2 (report) · hard H6 (pillars)	Soft; removed only in Sprint mode, with consent
D4	Minimum sleep	≥ 6 h (report) · ≥ 7 h (pillars)	Student's window is the lock; warn below 7 h
D5	Priority source	High/Med/Low (report) · grade weight (pillars)	Ask "counts for" (grade weight → p) + override
D6	Urgency / score	imp × exp(−slack/24) · 0.5U + 0.3P, U = 1 − h/168	One formula in the product; the other in Lab
D7	Step vs. session	25–45-min micro-tasks · 45–90-min parts	**Step** = content; **session** = time piece (45–90 min)
D8	Slot context (place)	Hard rule H5 (pillars) · not yet (report)	Leave out of V1 (no data source)
D9	Deadline format	Weekday names (baseline schema)	ISO datetimes
D10	Current time	Implicit in urgency / slack	Explicit `now` input
D11	Learning loop	Automatic updates (Pillar 4 Stage 5)	Suggest → student confirms
D12	7-day peer study?	Planned (report §9.8)	Decide by **W3**, because it decides Phase 2
D13	AI required (R4)?	M1 planned W4–W5	If yes, M1 only fills the TaskSheet

7 · Next steps by week (UI track runs in parallel with the engine)

W1	Tokens + components; clickable wireframes; **agree the JSON contract** + mock data; freeze the rule sheet (D1–D11)
W2	Week view on mock data (FullCalendar); TaskSheet; Quick-add parser with unit tests; works at 375 px
W3	Connect `/api/plan` (EDF V0); proposal mode; Why here?; Lab skeleton + scenario loader · **decide D12**
W4	Overload panel; pin; drag + validate-move; `.ics` export
W5	**Usability test #1** (5 non-team classmates, think-aloud, SUS) → fix the top 5 issues · M1 output → TaskSheet
W6	Something came up + change review + Undo (with M4); scored-greedy reasons (energy, order, break)
W7	Onboarding (3 questions, `.ics` import); progress update · if D12 = yes: Today + check-in + PWA
W8–9	**Feature freeze at start of W8** · study build · Lab compare view · figures from real runs only · bug fixes
W10	Demo: 3-min student story (setup → plan → why → disruption → overload) + 2-min Lab; rehearse twice

8 · UI ↔ engine contract (agree in W1)

- Times are **ISO 8601 with +07:00**; the engine converts them to slots; due after the week → slot ≥ 672.
- **`now` is an explicit input**: same input + same `now` → same output.
- Every block carries `reasons[]`; the UI never invents an explanation.
- Block `state`: PROPOSED · ACCEPTED · PINNED · DONE.
- Storage: browser + JSON backup (Phase 1); SQLite keyed by an anonymous code (Phase 2).

`POST /api/plan` → status, blocks, unplaced, overload, metrics
`POST /api/repair` (LOCAL | REBUILD) → blocks, changes, stability
`POST /api/validate-move` → ok, violations
`POST /api/progress` → re-plan of remaining work
`POST /api/overload/preview` → blocks, changes
`GET /api/scenarios[/id]` · `POST /api/export/ics`

Reason codes: `BEFORE_DEADLINE` `EARLIEST_FREE_GAP` `BEST_FIT_GAP` `ENERGY_MATCH` `AFTER_PREREQ` `BREAK_AFTER` `ORDER` `MOVED_BY_REPAIR` `PINNED_BY_USER`

Stack (keeps the baseline's no-Node.js choice): FastAPI + static web · **FullCalendar v6** (MIT; 15-min slots, `slotMinTime/MaxTime` hides sleep, `eventAllow` for live validity) · Alpine.js or plain JS · one `tokens.css` · Lucide · Inter · Python `icalendar` · PWA in Phase 2. React + Vite only if 2+ members already know it; don't mix stacks.

9 · Definition of done for the MVP

- A new user reaches a first plan in **under 3 minutes** (checked with 5 classmates).
- Every placed block has **≥ 1 reason** from the engine.
- Every automatic change has a **preview and Undo**; pinned blocks never move.
- T01–T10 run from Lab with **0 hard violations**; same input + `now` → same output hash.
- Works at **375 px** and on desktop; WCAG AA contrast; usable by keyboard.
- No placeholder numbers in slides: **every figure comes from a Lab run**.